

Quant Business Models and Roles

Giuseppe A. Paleologo

FINC-B8420 — Quantitative Investing

Lecture 1

What Is This Course?

- ▶ A rigorous, hands-on introduction to quantitative investment strategies
 - ▶ factor models, portfolio construction, risk management, performance attribution
- ▶ Targeted at MBA students with STEM background planning to work in asset management
- ▶ Our hope is that this course, combined with a a more traditional asset-management discipline, will be useful
 - ▶ VI, PE/VC, RE, ...
- ▶ Workload will be high.
- ▶ **Warning:** new course
 - ▶ Lectures, HWs are all new and will have problems
 - ▶ Please provide feedback, ask questions, interact.

Prerequisites

- ▶ You should already know:
 - ▶ linear algebra and basic optimization.
 - ▶ basic Financial markets (CMI)
 - ▶ Probability and statistics
 - ▶ Python and SQL
 - ▶ use AI tools intelligently, debug, and explain your own code
- ▶ Everyone should have completed self-assessment.
 - ▶ If you missed more than a few questions, and didn't do further review, the course will be close to impossible.

Quantitative Orientation

- ▶ This is a quantitative, hands-on course
 - ▶ build and test quant strategies in 5 HWs and final project.
 - ▶ *databases*: CRSP, TAQ, Compustat, Datastream (via WRDS)
 - ▶ use Python and SQL
 - ▶ use risk models and optimizers
 - ▶ use coding tools (Claude Code, Codex, ...)
- ▶ Guest speakers from leading quant firms to complement lectures
- ▶ *Final project*: an original strategy, backtested out-of-sample

Readings

1. Textbook: Giuseppe A. Paleologo, *The Elements of Quantitative Investing* (Wiley, 2025)
2. Academic/practitioner papers, distributed via Canvas
 - ▶ typically 1–2 articles per week
3. Optional (but useful) texts:
 - ▶ Paleologo, *Advanced Portfolio Management: A Quant's Guide for Fundamental Investors* (Wiley, 2022)
 - ▶ Pedersen, *Efficiently Inefficient* (Princeton, 2015)



Canvas

- ▶ All materials — syllabus, lecture notes, problem sets, readings — posted on Canvas
 - ▶ courseworks2.columbia.edu
- ▶ Submit problem sets and the final project via Canvas
- ▶ Communications from instructors and TAs are primarily through Canvas
 - ▶ check regularly for announcements and messaging notifications

Course Outline

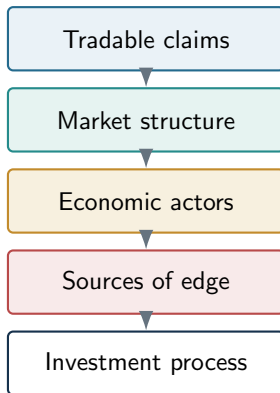
- ▶ 1. Course Introduction; Quant Business Models and Roles
- ▶ 2. Portfolio Construction Basics; Time-Series and Cross-Section tests
- ▶ 3–5 μ Models
 - ▶ factor models; anomalies; momentum and conditional strategies
- ▶ 6–7. Σ Models & t-costs
 - ▶ statistical factor models; price impact
- ▶ 8–9 Optimization
 - ▶ Static & dynamic optimization; constraints
- ▶ 10. Performance Attribution; Special Topics
- ▶ 11–12. Student Presentations
 - ▶ groups present their quantitative strategy, out-of-sample backtest results, risk and transaction-cost model analysis, and key findings
 - ▶ final project writeup due no later than one week after Session 12

Goals

- ▶ **Construct and estimate** cross-sectional factor models: fundamental, time-series, statistical
- ▶ **Implement** mean-variance portfolio optimization, and understand its practical limitations
- ▶ **Build and apply** risk models: BARRA-style factor models, shrinkage-based covariance estimators
- ▶ **Design and backtest** strategies (value, momentum, quality, conditional variants), accounting for transaction costs and data biases
- ▶ **Incorporate transaction costs** into optimization, myopic and dynamic
- ▶ **Evaluate performance**: Sharpe/information ratios, factor attribution, transfer coefficients
- ▶ **Identify and mitigate** pitfalls: look-ahead bias, data mining, overfitting, crowding

What, Where, Who, How

1. Which securities can quantitative strategies trade?
2. Where and how do those securities trade?
3. Who supplies liquidity, demands it, and bears risk?
4. Why can excess returns survive competition?
5. How does a quantitative manager turn data into PnL?



Quantitative Strategies

- ▶ A better label would be *systematic* strategies
- ▶ Distinguishing feature:

large number of decisions, over time and across assets

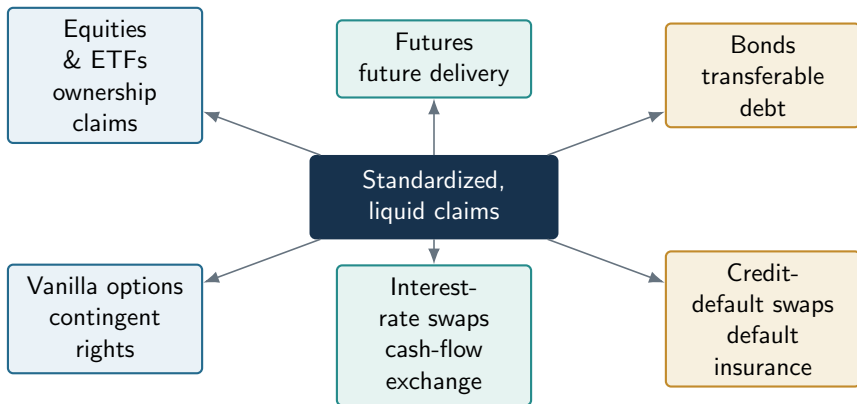
Consequences:

1. relatively low payoff per investment decision
 2. automation of decisions
 3. large cross-section of investable assets and/or high frequency of decisions
- ▶ We make no claims about the source of “alpha”, or the asset class

A Very Short History

- ▶ 1960s: Ed Thorp (derivatives arbitrage, early factor investing, converts, cointegrated pairs)
- ▶ 1970s: Index investing (Wells Fargo)
- ▶ 1980s: early factor investing (and associated literature), institutional, buy/sell side proprietary, hedge funds
- ▶ 1989: Renaissance Technologies (futures stat arb, equities stat arb)
- ▶ Late 1990: prop trading desks, within investment banks and outside
- ▶ 2000s: hedge fund growth, specialized and platforms
- ▶ 2008: beginning of the end of bank prop trading desks. Spin-offs: Lehmann-Barclays-Squarepoint; Deutsche-GSA; Morgan Stanley-PDT
- ▶ Post 2008: consolidation: rise of the HF platforms and prop trading firms; growth of passive management and decline of institutional active management
- ▶ Mid 2020s: AI reshapes short-horizon strategies and long-term investment processes

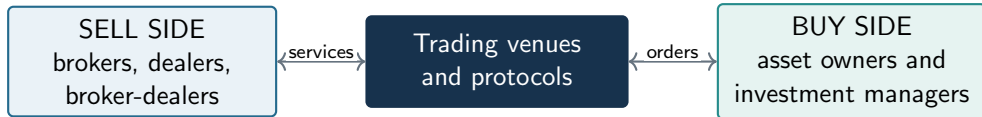
Securities



Three Venue Types

	Exchange	OTC	Dark pool
Matching	Central order book	Bilateral or dealer network	Alternative trading system
Visibility	Quotes and orders visible	Dealer quotes; often opaque	Orders hidden pre-trade
Typical assets	Stocks, options, futures	Bonds, swaps, FX, CDS	Mostly equities
Core benefit	Competition and immediacy	Flexibility and dealer intermediation	Reduced information leakage
Main concern	Fragmentation and fees	Counterparty and price discovery	Limited transparency

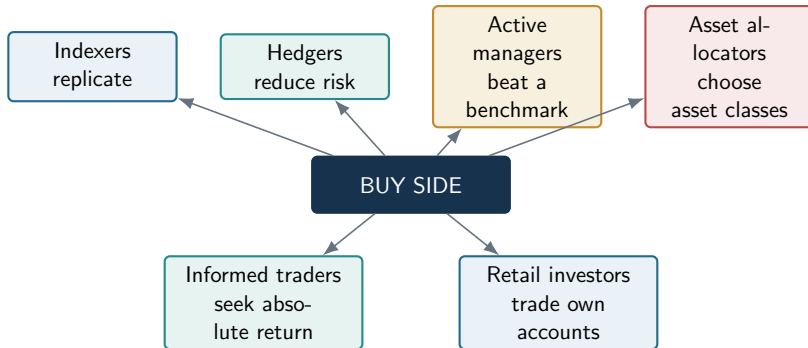
Markets connect liquidity providers with investors who need trades



Sell side: facilitates transactions and earns spreads, commissions, financing income, and service fees.

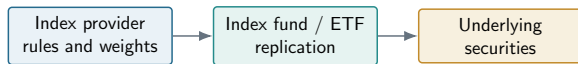
Buy side: trades for beneficial owners, accepting risk in pursuit of investment or hedging objectives.

The buy side is best understood by investment objective



Source: Classification used in the chapter; it is organized by economic function rather than legal form.

Indexers accept market returns and create predictable flows



- ▶ Objective: minimize tracking error and cost.
- ▶ Reconstitution dates are scheduled.
- ▶ Additions, deletions, and weight changes force trades.
- ▶ The mechanical demand is valuable information to liquidity providers.

Active managers and allocators solve different benchmark problems

Institutional active manager

Portfolio = benchmark + tracking portfolio

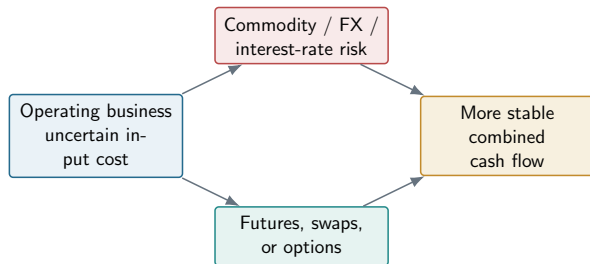
- ▶ seeks benchmark-relative return;
- ▶ tracking error limits discretion;
- ▶ very low tracking error can produce “closet indexing.”

Asset allocator

Portfolio = weighted asset classes

- ▶ chooses exposures to equities, bonds, commodities, and cash;
- ▶ may allocate to private equity, venture capital, hedge funds, and real estate;
- ▶ changes weights relatively slowly.

Hedgers trade to reduce risks created by their core business



The hedge is not required to have attractive standalone expected return. Its role is to offset an exposure elsewhere in the firm.

Typical examples include airlines hedging fuel and exporters hedging foreign-currency revenue.

Informed traders improve prices and bridge future liquidity needs

Price discovery

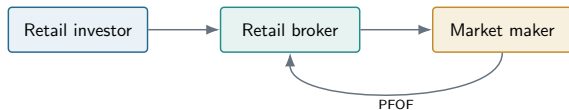
1. estimate a security's value;
2. compare value with market price;
3. trade against the mispricing;
4. move price toward the estimate.

Liquidity provisioning

1. forecast a future imbalance;
2. establish inventory in advance;
3. absorb demand at the event;
4. earn compensation for risk and immediacy.

The forecast horizon can range from sub-second market making to multi-week event strategies.

Retail flow is valuable partly because it is less informed

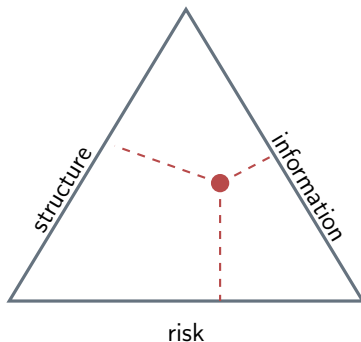


- ▶ The market maker earns the spread and faces relatively low adverse-selection risk.
- ▶ The broker receives compensation for routing.
- ▶ Best execution remains the central client-protection question.

Source: The chapter cites Barber and Odean's evidence on retail-trading performance.

The Sources of Alpha

I find it useful to think about alpha as being a combination of three idealized mechanisms: **information** (“investing”), **structure** (“trading”) and **risk** (“risk premia”).



Investing

“A capital market is said to be **efficient** if it fully and correctly reveals all available information in determining security prices. Formally, the market is said to be efficient with respect to some information set, ϕ , if security prices would be unaffected by revealing that information to all participants. Moreover, efficiency with respect to an information set, ϕ , implies that it is impossible to make economic profits by trading on the basis of ϕ .” (Malkiel)

This is the definition of **investing alpha**. It's not covering all that we refer to as alphas.

Investing is Trading Based on Differential Information

St. Thomas Aquinas gives the example of a merchant who knows a ship with grain is on the way to a famine-stricken city. The question: can the merchant sell his grain at the current high price without revealing the fact that soon there will be more grain and lower prices?

He answers that yes, it is ethical. We also say that the merchant has investment alpha. We also say that, by trading, we reveal private information and perform price discovery (which is a good thing).

Features of Investing Strategies

1. Trading is costly, but it is not a source of profit in the sense that we would not trade, had we not differential information. We try to minimize costs, while expressing information.
2. Investing can occur both at very long horizons and very short ones .
3. “Holding ideally forever” or “focusing on the fundamental value” of a security. These notions of investing are ill-defined. You have a) a differentiated thesis vs consensus reflected by current prices, and b) an investment horizon .

Examples

None of these investment styles are purely investing. But the vast majority of their PnL derives from time-varying information.

1. Long/Short equities
2. Systematic equities and credit
3. Front-end liquid futures
4. Long-Term Commodities Futures
5. Open and Closing Options
6. Market-Making Crossing Strategies

Trading

Let's change Thomas' example a bit. Every Spring the city (ancient Rome) holds a harvest festival, and consumes an unduly amount of honey. He knows the travel routes to Hispania, where honey is produced. He buys from producers there, and ships to the city. Where he makes a profit.

The information about demand is public. He combines that by taking advantage of market structure and access, and costs.

This is trading.

Features of Trading Strategies

1. Trading is costly, and it is the source of profit.
2. There is an information advantage, but it relates to market structure and participant preferences .
3. Execution, market access, funding, regulations are all essential features of the strategies.
4. A few basic mechanisms:
 - (a) Violation of the law of one price, due to market structure or funding
 - (b) Heterogenous liquidity (time) preferences
 - (c) Heterogeneous funding status

Trading Relies on Participants Heterogeneous Preferences, and Market Structure

Modes of exchange:

- ▶ Trading exchanges
- ▶ OTC
- ▶ Dark pools

Participants:

- ▶ Sell-Side:
 - ▶ Brokers
 - ▶ Dealers
 - ▶ Broker-dealers
- ▶ Buy-side:
 - ▶ Indexers; Hedgers; Institutional managers
 - ▶ Asset allocators; Retailer traders; Informed Investors

Reference (a bit dated): Harris, *Trading & Exchanges*; EQI

Trading Styles Examples

Trade	Edge	Cadence
Index Rebalancing	structural flow predictability	3/6/12 months
Exchange for Physical	funding/liquidity/law of one price	1/3 months
Swap & Treasury Bond Basis	liquidity	event-driven
ADR & ETF Arbitrage	liquidity/law of one price	event-driven
Bond Auctions	liquidity	monthly
Closed-End Funds	structural/law of one price	monthly
Carry Trades	liquidity	event-driven
End-of-Month funsa	flow predictability	monthly
SPACs	liquidity/structural	monthly
Market Making Add	liquidity/structural	intraday
IPOs and Secondary	liquidity/structural	event-driven
Conversion Trades	funding	weekly
DeFi	structural	intraday
Delta One	funding/structural	event-driven
Merger Arb	liquidity/law of one price	event-driven
Convertible Bonds	liquidity	event-driven

Risk Premia

You are asked by your investors to outperform the risk-free rate.

You invest in a market index (IVV, IWM, EFA), but you may also invest in asset-specific benchmarks (SGOV, TLT), sectors (XLE, XLT, XLF...), “investment styles” (VLU, SZ, MTUM, QUAL, MVOL)...

These portfolio returns should outperform a benchmark, if chosen well.

In any event: you collect fees. You play golf during the weekend. American Dream.

Risk premia compensate investors for losses in bad times

Let R_i denote a gross return and let m be the stochastic discount factor. m is high in states in which an extra dollar is especially valuable.

$$1 = \mathbb{E}[mR_i] = \mathbb{E}[m]\mathbb{E}[R_i] + \text{Cov}(m, R_i), \quad R_f = \frac{1}{\mathbb{E}[m]},$$

$$\Rightarrow \mathbb{E}[R_i] - R_f = -\frac{\text{Cov}(m, R_i)}{\mathbb{E}[m]}.$$

Positive risk premium

$\text{Cov}(m, R_i) < 0$: the asset performs poorly in bad times, when wealth is most valuable.
Investors demand a higher expected return.

Insurance-like asset

$\text{Cov}(m, R_i) > 0$: the asset pays in bad times.
Its hedging value allows a lower expected return.

Factor exposures turn risk compensation into a measurable model

Approximate the discount factor with centered systematic factors f :

$$m = a - b^\top f, \quad \mathbb{E}[f] = 0.$$

Then

$$\mathbb{E}[R_i] - R_f = R_f b^\top \text{Cov}(f, R_i) = \underbrace{\beta_i^\top}_{\text{quantity of factor risk}} \underbrace{\lambda}_{\text{price of factor risk}},$$

where

$$\beta_i = \text{Var}(f)^{-1} \text{Cov}(f, R_i), \quad \lambda = R_f \text{Var}(f) b.$$

Expected excess return is compensation for systematic exposure; diversification removes idiosyncratic risk, not factor risk.

Factor investors hold portfolios, not abstract labels

- ▶ **Market:** risky assets rather than cash
- ▶ **Value:** cheap rather than expensive securities
- ▶ **Size:** small rather than large firms
- ▶ **Momentum:** recent winners rather than losers
- ▶ **Quality:** profitable, robust firms rather than weak firms
- ▶ **Low volatility:** defensive rather than high-beta stocks
- ▶ **Carry:** high-yielding rather than low-yielding assets

Economic interpretation requires care

A persistent factor return may reflect compensation for bad-state exposure, investor behavior, or institutional constraints. Calling it a “risk premium” does not establish which mechanism is responsible.

Information and Constraints

A unified interpretation: **alphas come from investor heterogeneity**.

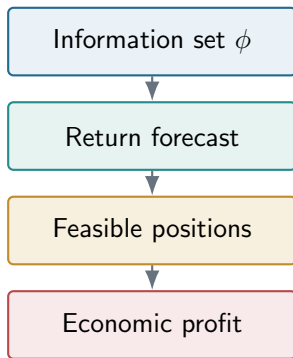
1. Liquidity constraints
2. Funding constraints
3. Flow constraints
4. Risk constraints
5. Informational advantage

- ▶ Usually researchers concentrate on the latter, which is an epiphenomenon of the former. Great traders always ask themselves who is on the other side of the trade.
- ▶ With the popularization of black boxes, intelligence (in the original sense of the word, *inter+legere*, “read into [something]”) will be in even higher demand.
- ▶ Keep in mind this statement by Cochrane, first-order correct: “There is no alpha. There is the beta you understand, and the beta you don’t understand”.

Market efficiency depends on both information and implementation

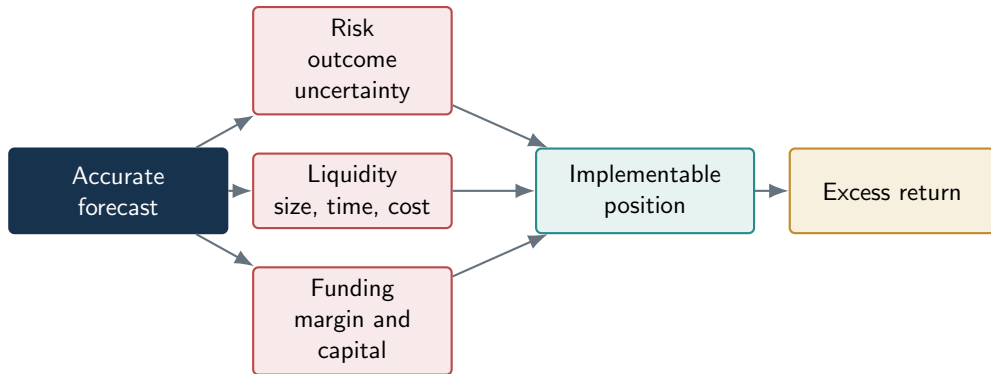
Efficiency with respect to ϕ

Prices incorporate ϕ when revealing it to everyone would neither change prices nor support economic profits.



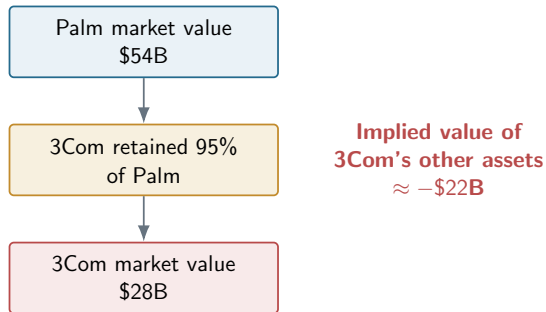
Predictability alone does not guarantee profit: the forecast-to-position path can fail.

A correct forecast may still be impossible to monetize



A constraint borne by one investor may be the edge earned by another.

The Palm–3Com case shows how a visible arbitrage can survive

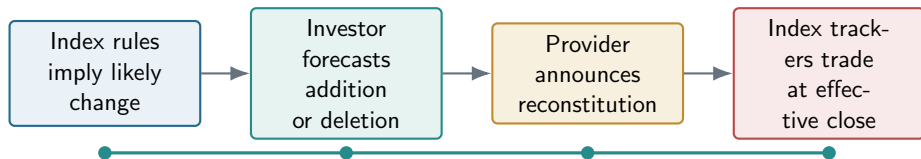


The apparent trade was to buy 3Com and short Palm.

But only 5% of Palm had been floated. Palm shares were hard or extremely expensive to borrow, so the short leg constrained the arbitrage.

Lesson: a valuation identity is not a trade until every leg can be financed and executed.

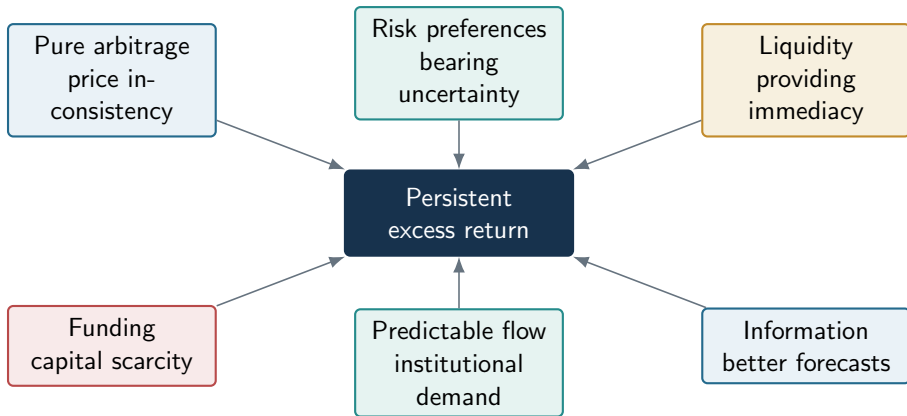
Predictable institutional flows create event-driven opportunities



Liquidity providers may buy additions or sell deletions before the event, then unwind into benchmark-tracker demand. They bear market, company, and event-cancellation risk while waiting.

Source: Chapter example: Tesla's addition to the NASDAQ-100 in July 2013.

Six mechanisms explain why excess returns can persist



Quantitative investing begins with four families of data

Prices & volume

Regular bars or order-level exchange data.

Characteristics

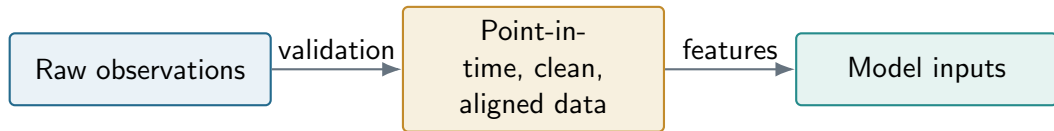
Security-level descriptors stamped by time.

Time series

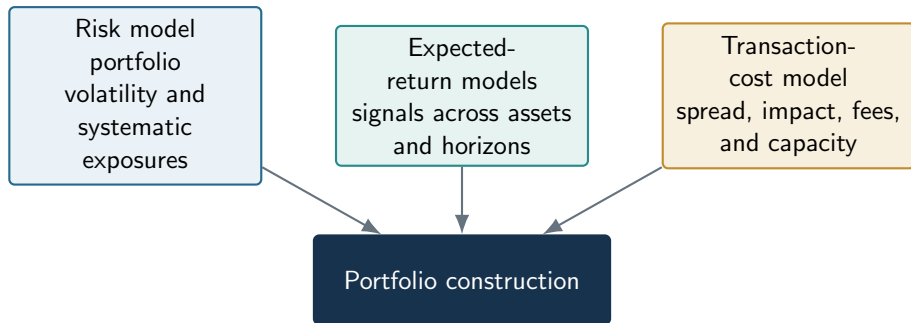
Macro and market variables not tied to one security.

Unstructured

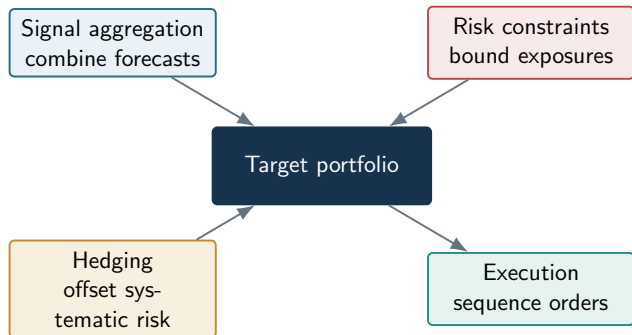
Text, images, audio, video, and multimodal records.



Before the trade, three models define the opportunity set



During the trade, the system converts estimates into positions

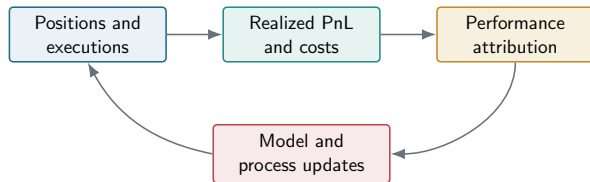


A typical objective balances forecast return against risk and trading cost:

$$\begin{aligned} \max_w \quad & \hat{\mu}^\top w \\ & - \lambda w^\top \Sigma w \\ & - C(w - w_0). \end{aligned}$$

The optimizer is a decision rule, not merely a mathematical exercise.

After the trade, realized PnL becomes new information



- ▶ Which signals worked?
- ▶ Did sizing add value?
- ▶ Were realized costs consistent with the model?
- ▶ How should risk and leverage change through time?